Ashima Sharma
Software Engineer

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Summary

Software Engineer with 5+ years of experience designing and building scalable full-stack, cloud-native, and AI-driven applications using Python, TypeScript, Node.js, and React. Proven expertise in developing microservices architectures, distributed systems, real-time platforms, and LLM-powered applications using RAG pipelines, deployed across AWS and Google Cloud environments. Strong background in API design, event-driven systems, CI/CD automation, and performance optimization, with experience delivering production-scale enterprise platforms in Agile engineering teams.

Skills

Programming Languages: Java, Python, JavaScript (ES6+), TypeScript, SQL, Bash
Frontend Development: React.js, Redux, HTML5, CSS3, Bootstrap, Responsive UI Development, WebSockets
Backend & APIs: Node.js, Express.js, Django, Django REST Framework, FastAPI, Flask, REST API Development, Microservices Architecture, API Security (JWT, OAuth2, RBAC, SSO)
AI / ML / LLM Systems: LLM Applications, LLM Agents, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search (FAISS), Vector Databases, scikit-learn, Pandas, NumPy, NLP Pipelines, Model Training & Evaluation
AI-Assisted Development Tools: Claude Code, Cursor
Cloud Platforms: AWS (EC2, S3, RDS, Lambda, IAM), Google Cloud Platform (Cloud Run, Pub/Sub, Cloud Storage), Multi-Cloud Deployments
DevOps & Infrastructure: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, AWS CodePipeline, CI/CD Automation, Infrastructure as Code
Databases & Distributed Systems: PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka, Database Schema Design, Data Modeling, Event-Driven Architecture, Asynchronous Processing, Distributed Caching, Celery
Data Visualization & BI: Tableau, KPI Dashboards, real-time analytics dashboards
Testing, Observability & Practices: Unit Testing, Integration Testing, End-to-End (E2E) Testing, JUnit, Jest, Postman, Grafana, ELK Stack, Prometheus, Agile/Scrum, System Design, Performance Optimization

Experience

Software Engineer | Metis — **Jul 2025 – April 2026**

- Built a multi-tenant manufacturing workflow production platform MVP from 0 to 1, using Flask, React + JavaScript UI with reusable component architecture, Redux state management, and optimized rendering via code-splitting and lazy loading.
- Designed and implemented Python (Flask) based REST APIs and microservices with authentication, input validation, error handling, and PostgreSQL integrations supporting enterprise workflows.
- Designed and optimized PostgreSQL database schemas supporting multi-tenant enterprise workflows, ensuring data integrity, scalability, and efficient query performance.
- Developed LLM-powered AI report generation services using Retrieval-Augmented Generation (RAG) pipelines to retrieve contextual enterprise data and automate structured reporting.
- Integrated Google Vertex AI vector search with RAG pipelines for semantic retrieval, enabling contextual document search and intelligent data access across equipment manuals and enterprise knowledge bases.
- Engineered data ingestion and indexing pipelines extracting structured JSON using Gemini models, storing artifacts in Google Cloud Storage, and indexing content for semantic retrieval.
- Architected event-driven distributed workflows using Google Cloud Pub/Sub enabling asynchronous processing across scalable microservices.
- Implemented enterprise Single Sign-On (SSO) and Role-Based Access Control (RBAC) using WorkOS, enabling secure multi-tenant authentication and fine-grained permissions management.
- Containerized and deployed microservices using Docker, Google Cloud Run, Container Registry, and CI/CD pipelines, integrating end-to-end workflow testing to prevent regressions and improve release reliability.
- Leveraged AI-assisted development tools (Claude Code, Cursor) to accelerate full-stack feature delivery, improve code quality, and streamline development workflows.

Researcher I — X-Lab, University of Alabama at Birmingham — ***Publication*** — **May 2025 – Jul 2025**

- Developed finite element simulation datasets and trained RandomForest (scikit-learn) predictive models to

analyze stress-strain behavior in soft robotics systems.
- Automated data preprocessing, feature engineering, and model training pipelines using Python (NumPy, Pandas), improving experimentation and iteration speed by 40%.
- Optimized feature selection and hyperparameter tuning to enhance prediction accuracy across experimental datasets.
- Validated machine learning models against structured test datasets, ensuring performance reliability and reproducibility.
- Collaborated within Agile research teams to refine algorithms, monitor model performance, and improve experimentation and deployment processes.

Software Engineer | Cybhop Tech **Apr 2023 – Feb 2024**
- Developed scalable B2B enterprise applications using Python (Django, FastAPI, Flask), TypeScript and React microservices-based, delivering robust REST API backend services and responsive frontend interfaces.
- Designed and implemented secure JWT/OAuth2 authentication and RBAC authorization systems supporting multi-service environments and protecting access for 50k+ users.
- Built responsive React + Redux user interfaces improving user experience and reducing page load times by 40% through frontend optimization.
- Designed scalable SQL and NoSQL database schemas (PostgreSQL, MySQL, MongoDB) through schema design and query tuning, improving retrieval performance by 30%.
- Implemented asynchronous processing pipelines using Celery, Pandas, and NumPy, increasing backend throughput and large-scale data processing efficiency.
- Deployed containerized microservices using Docker, Kubernetes, AWS EC2, S3, RDS, Lambda, Jenkins, and AWS CodePipeline, enabling automated CI/CD releases and scalable event-driven architectures.
- Conducted peer code reviews and mentored junior engineers, improving code quality and adherence to engineering best practices.

Software Engineer | Taboola **Mar 2020 – Mar 2023**
- Designed and developed large-scale full-stack AdTech platform features using Python (Django), Node.js, and React.js, supporting content recommendation, campaign management, and publisher monetization workflows.
- Engineered Kafka-based event-driven pipelines to process high-volume AdTech events in real time, improving asynchronous communication between recommendation, campaign management, and publisher monetization microservices.
- Built automated alerting systems integrated with Slack and email to detect anomalies and threshold breaches (e.g., publisher revenue drops, traffic anomalies, system latency spikes), improving proactive incident response.
- Improved REST API performance by 40% through Redis caching, asynchronous processing, and query optimization, enhancing application responsiveness and scalability.
- Developed Tableau-based analytics dashboards and reporting systems for ad performance, campaign KPIs, and system health monitoring enabling real-time decision-making across business teams.
- Migrated legacy .NET monolithic modules to microservices-based architecture, enabling phased modernization, scalability, and independent service deployments.
- Enhanced system observability using Grafana, ELK Stack (Elasticsearch, Logstash, Kibana) and Prometheus, reducing debugging and incident resolution time by 30%.
- Designed, deployed, and managed AWS cloud infrastructure (EC2, S3, RDS, Lambda) with automated CI/CD pipelines using Jenkins and AWS CodePipeline, improving deployment efficiency.
- Implemented comprehensive unit, integration, and end-to-end testing frameworks using JUnit, Jest, Postman improving release reliability.
- Collaborated with cross-functional teams including product, data, and infrastructure to gather requirements and deliver full-stack platform features at scale.

Education

Master in Computer Science, *University of Alabama at Birmingham* Aug 2023 – May 2025
GPA: 3.8/4.0 **| Teaching Assistant:** Machine Learning (Aug-Dec 2024), Fundamentals of Data Science (Jan-May 2025)
Bachelor of Engineering in Computer Science, *Chandigarh University* Aug 2016 – May 2020
GPA: 7.03/10.0